

WEEK 8 PROJECT REPORT

STUDENT PERFORMANCE DASHBOARD

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Project Title: Student Performance DashBoard

Tool Used: Microsoft Power BI Desktop

Dataset: Students Performance Dataset

INTRODUCTION

The Week 8 task focused on developing an interactive **Power BI dashboard** using the Student Performance dataset. The purpose of this task was to visualize academic performance and analyze different factors such as subject scores, parental education, test preparation, lunch type, gender, and grades.

PREVIOUS WEEK – WEEK 7

In Week 7, the Student Performance dataset was cleaned and transformed using **Power Query**. Total Score and Average Score were calculated, followed by the creation of **Grade** and **Pass/Fail** columns using conditional logic. The data was then prepared for visualization and further analysis in Week 8.

DASHBOARD DEVELOPMENT

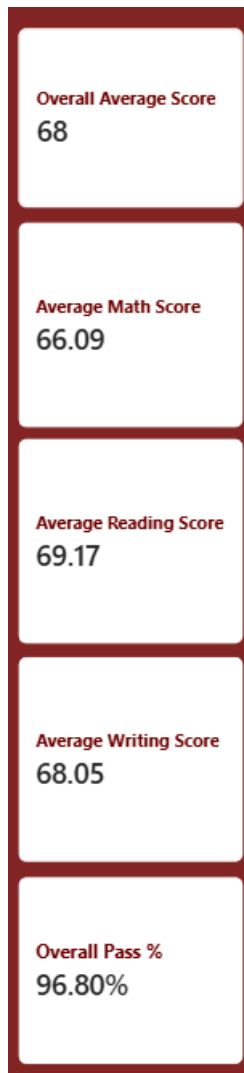
The Week 8 dashboard was designed to provide a clear overview of student performance. Different types of Power BI visuals were used to present the information in an easy-to-understand format.

KPI Cards

Five KPI cards were created to display the main performance indicators:

- **Overall Average Score:** 68
- **Average Math Score:** 66.09
- **Average Reading Score:** 69.17
- **Average Writing Score:** 68.05
- **Overall Pass Percentage:** 96.80%

These cards provide a quick summary of the overall academic performance.

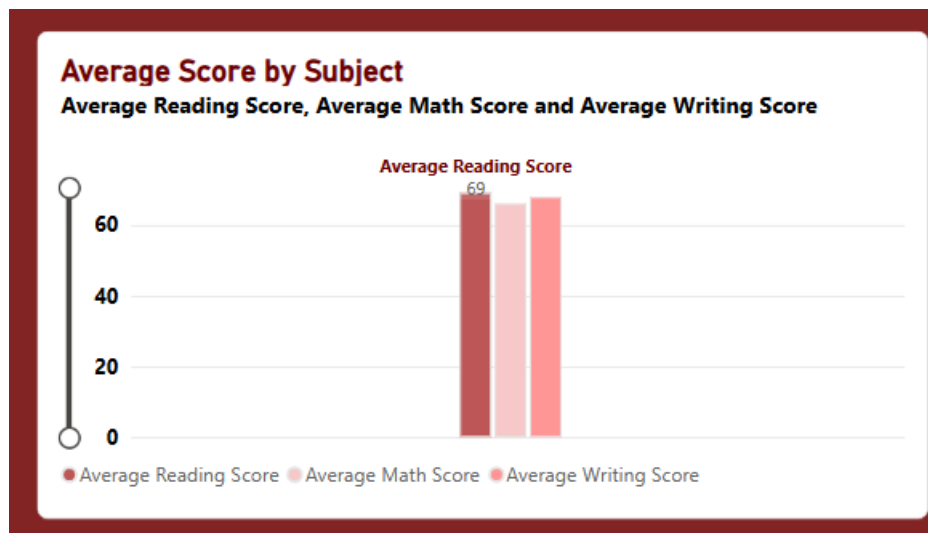


SUBJECT PERFORMANCE

A chart titled “**Average Score by Subject**” was created to compare the average marks obtained in Math, Reading, and Writing.

- **Math:** 66.09
- **Reading:** 69.17
- **Writing:** 68.05

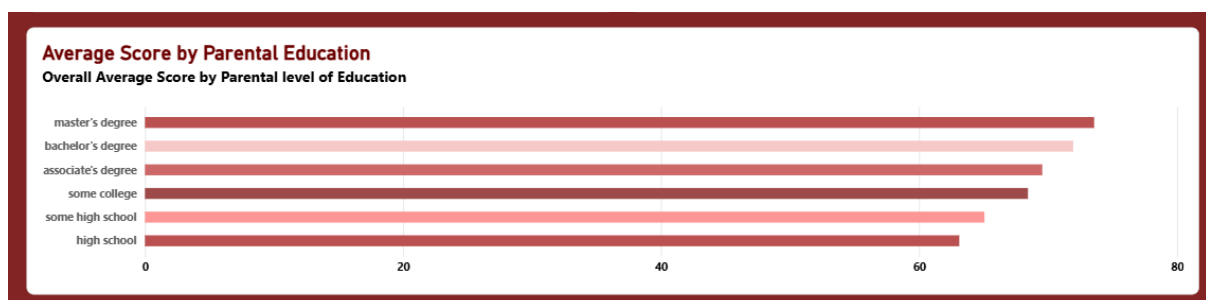
Reading had the highest average score, while Math had the lowest average among the three subjects.



PARENTAL EDUCATION ANALYSIS

A bar chart was created to analyze the relationship between **parental education level** and **students' average scores**.

The highest average score was observed for students whose parents had a **Master's degree**, while students whose parents had a **High School** education had the lowest average score among the categories.

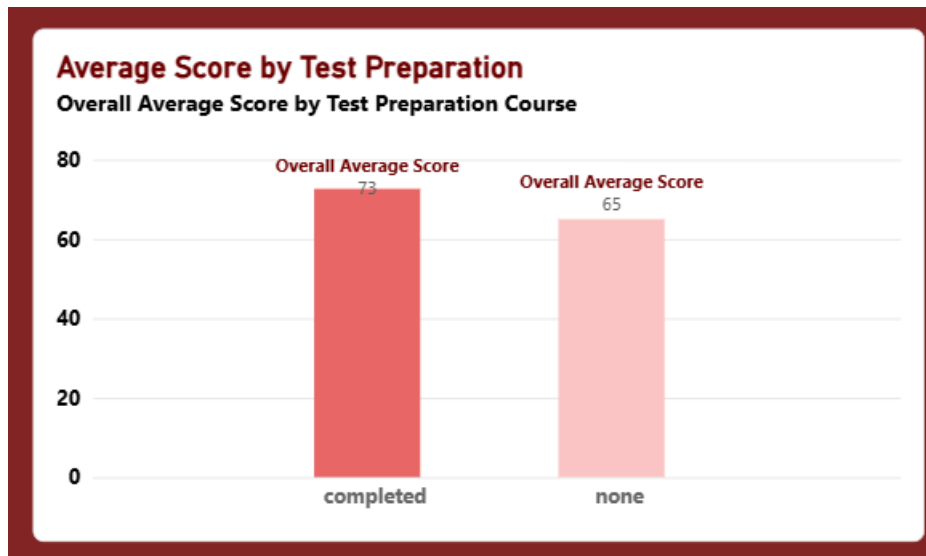


TEST PREPARATION ANALYSIS

A column chart was created to compare students who **completed** the test preparation course with those who did not.

- **Completed:** 73
- **None:** 65
- **Writing:** 68.05

The dashboard shows that students who completed the test preparation course had a higher average score. A **Test Preparation Course slicer** was also added to allow interactive filtering.

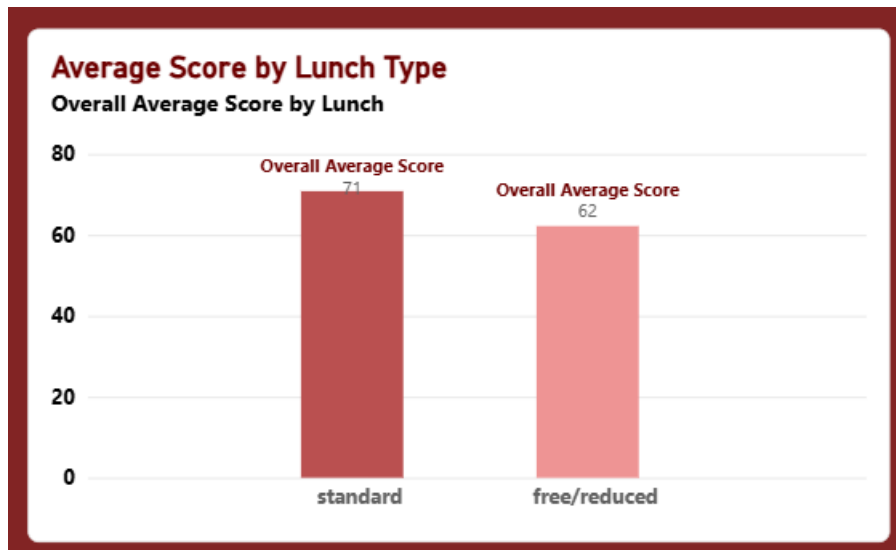


LUNCH TYPE ANALYSIS

The dashboard also analyzed average scores according to lunch type.

- **Standard:** 71
- **Free/Reduced:** 62

The Standard lunch group had a higher average score than the Free/Reduced group.



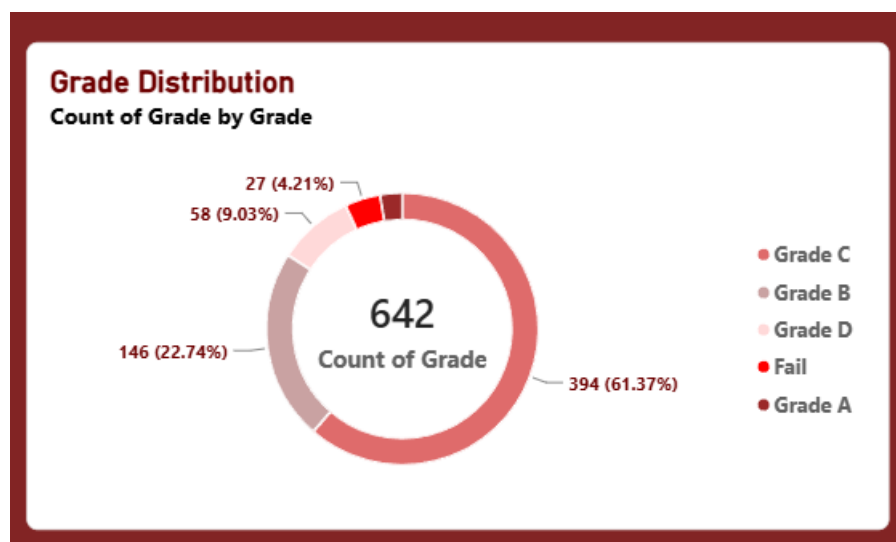
GRADE DISTRIBUTION

A donut chart was created to display the distribution of students according to their calculated grades.

The distribution was:

- **Grade C:** 573 students
- **Grade B:** 272 students
- **Grade D:** 73 students
- **Grade A:** 52 students
- **Fail:** 30 students

Grade C represented the largest group of students in the dataset.



INTERACTIVE SLICERS

The dashboard was made interactive by adding the following slicers:

- Gender
- Race/Ethnicity
- Parental Level of Education
- Lunch
- Test Preparation Course
- Subject

These slicers allow users to filter the dashboard and analyze specific groups of students.

Parental level of Education ☐ associate's degree ☐ bachelor's degree ☐ high school ☐ master's degree ☐ some college ☐ some high school	
Race/Ethnicity ☐ group A ☐ group B ☐ group C ☐ group D ☐ group E	
Test Preparation Course ☐ completed ☐ none	
Lunch ☐ free/reduced ☐ standard	
Gender ☐ female ☐ male	Subject ☐ Math Score ☐ Reading Score ☐ Writing Score

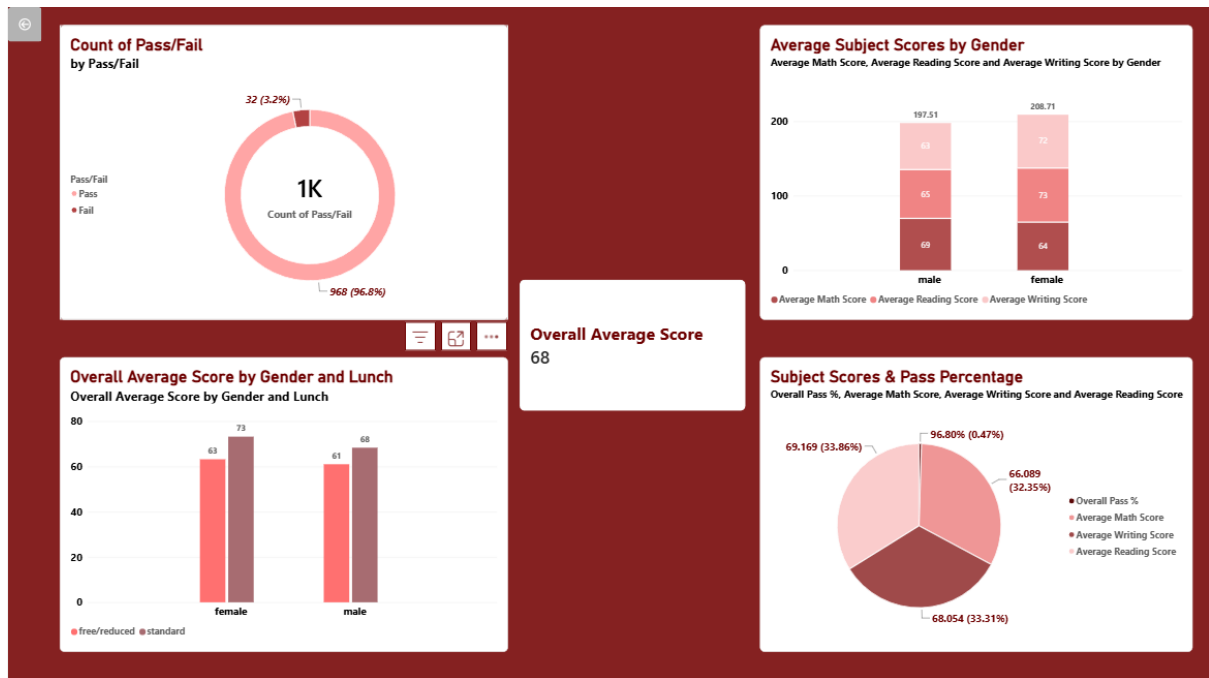
ACADEMIC PROFILE – DRILL-THROUGH

An **Academic Profile** page was created using the Power BI drill-through feature.

The page contains:

- Subject Scores and Pass Percentage
- Average Subject Scores by Gender
- Average Score by Gender and Lunch
- Pass/Fail Distribution
- Overall Average Score
- Back button for navigation

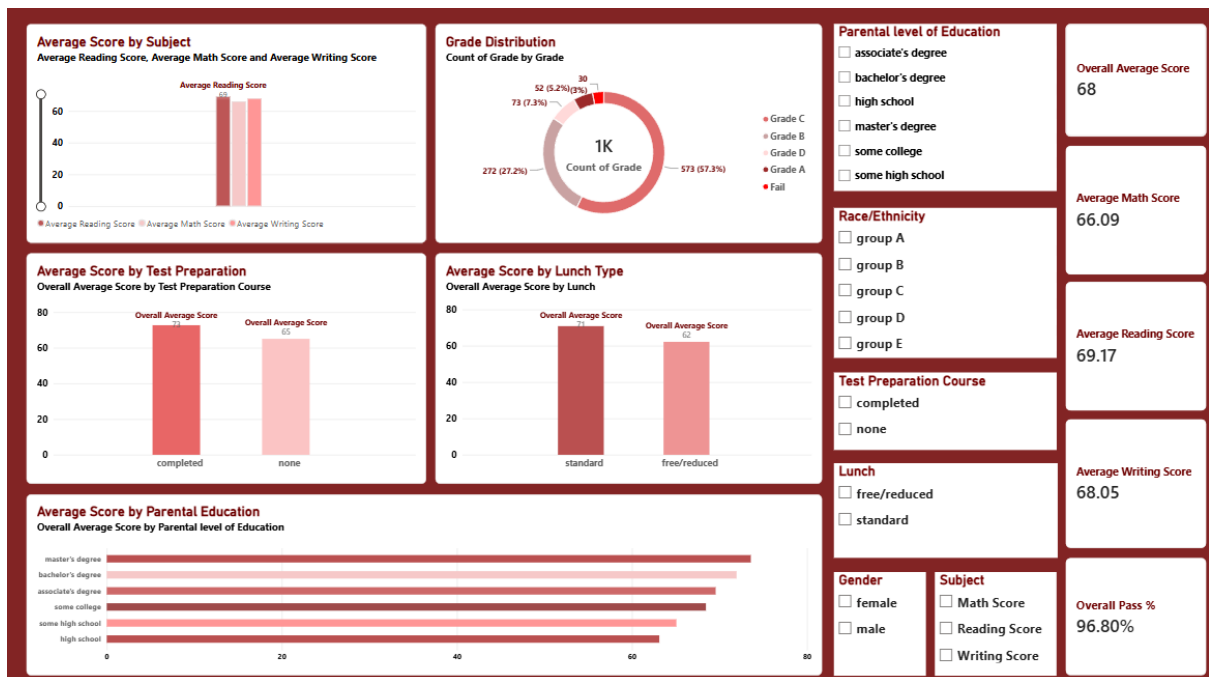
The **Gender** field was used as the drill-through field. This allows the user to move from the main dashboard to the Academic Profile for a more focused analysis.

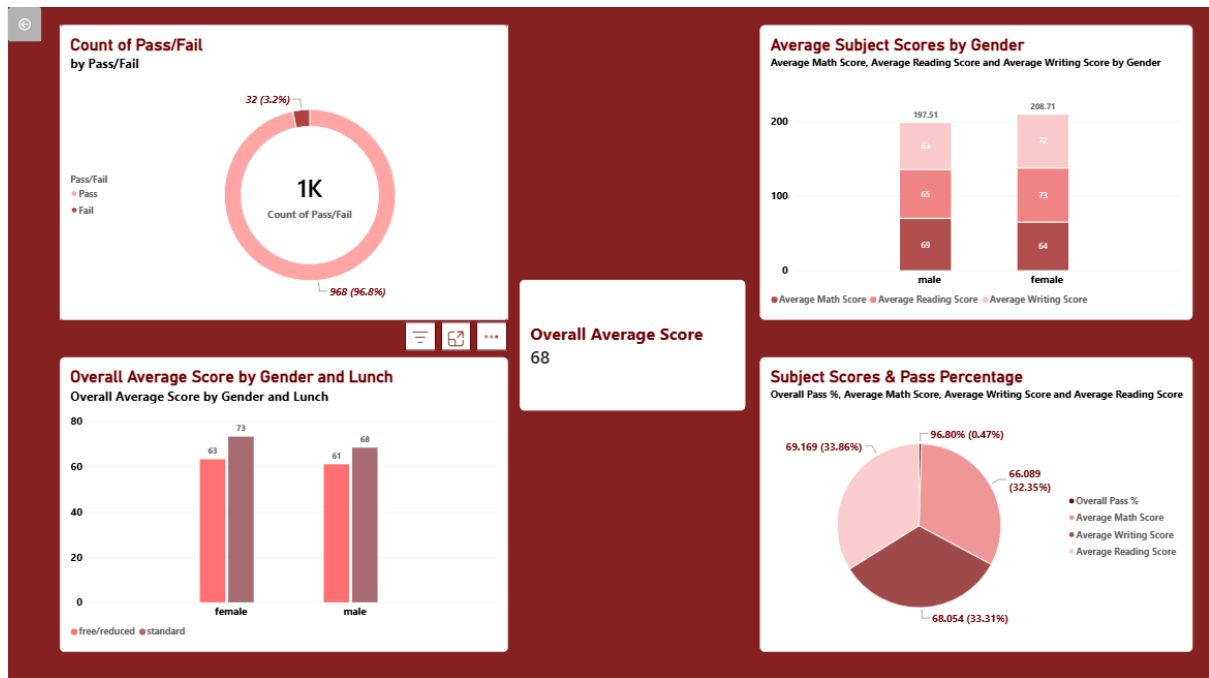


FINAL DASHBOARD

The final dashboard combines all the major analyses into an interactive Power BI report. It includes **KPI cards, charts, slicers, grade distribution, subject analysis, and drill-through functionality.**

The dashboard provides a simple way to explore student performance and compare different groups within the dataset.





CONCLUSION

The Week 8 task successfully completed the **visualization and dashboard development** stage of the Student Performance project. Power BI was used to convert the transformed Week 7 dataset into an interactive dashboard. The final dashboard provides useful insights into student performance across subjects and different demographic and educational factors. The use of **DAX measures, interactive slicers, KPI cards, and drill-through analysis** made the dashboard more informative and user-friendly.